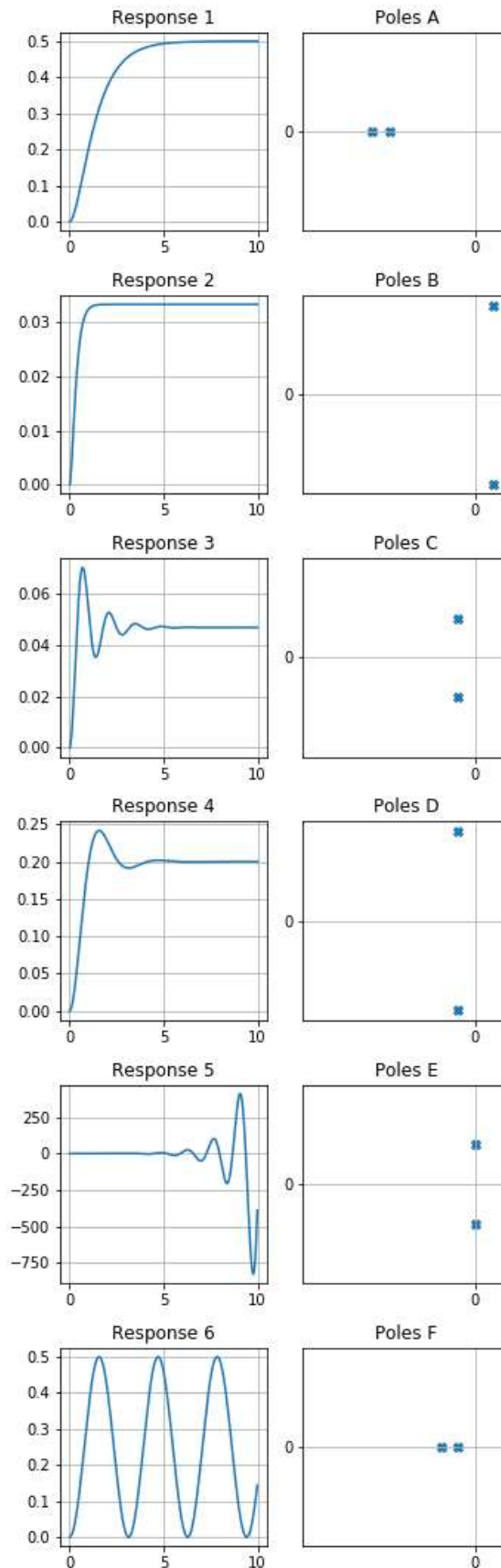


Tutorial 2: System Identification – Time Domain

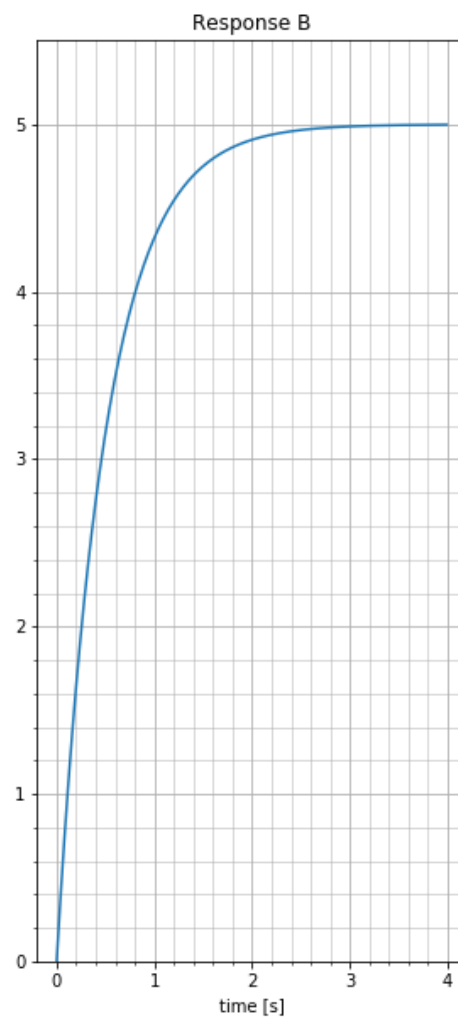
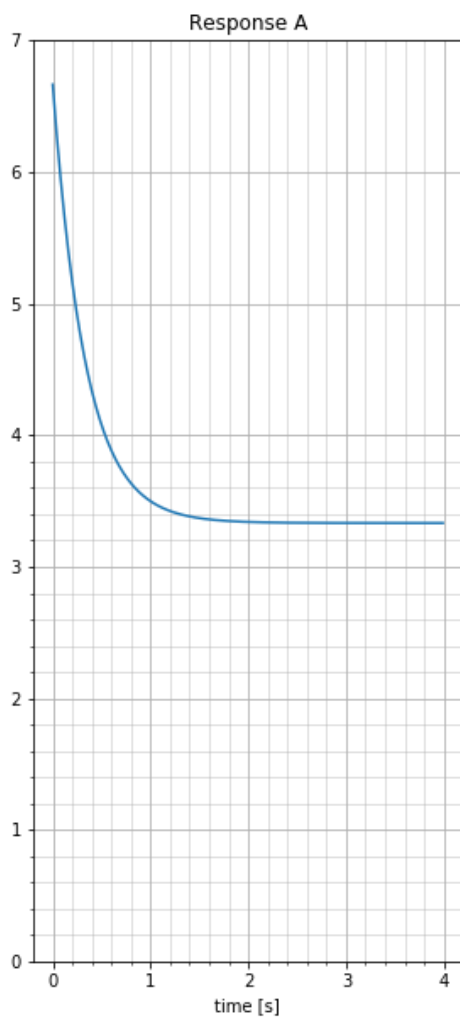
Question 1

Each of the following plots 1-6 represents the impulse response for a second-order system, while the plots A-F represent their pole locations. Match each response to the corresponding pole plot.



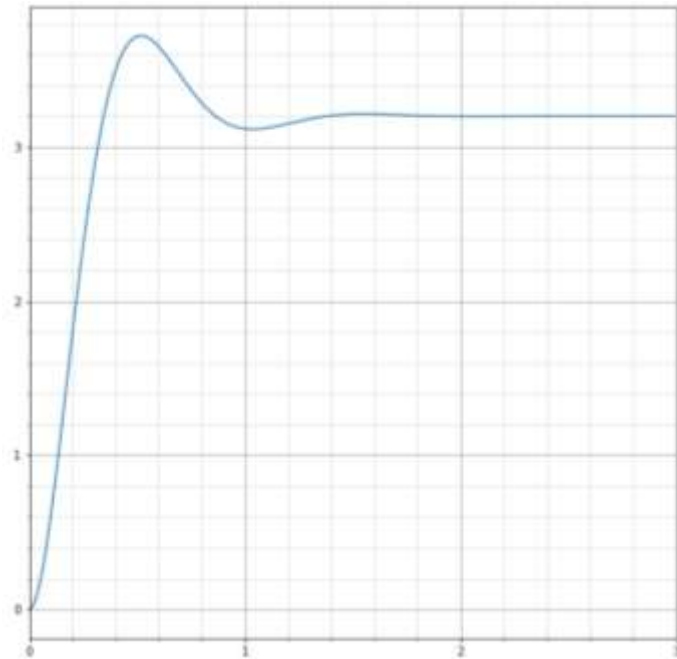
Question 2

The plots below show the step responses obtained from two unknown first-order systems. Suggest transfer functions for the systems based on these plots.



Question 3

- a) Find the transfer function of the second-order system that produces the following unit step response:



- b) The system is a spring mass damper with unit mass, and can be modelled by the differential equation

$$\ddot{x} = -b\dot{x} - kx + F$$

Identify the values of the spring and damping coefficients based on the transfer function.

Formulas:

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n}$$